

22	C	Based on the field pattern given, the smaller charge is on the left. Let x = distance between the neutral point and the smaller charge. At the neutral point (which is somewhere between the two charges), $E_1 = E_4$ $\frac{1}{4\pi\epsilon_0} \frac{(1.0\mu\text{C})}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{(4.0\mu\text{C})}{(6.0\text{ cm} - x)^2}$ $\frac{1}{x^2} = \frac{4}{(6.0 - x)^2} \Rightarrow \frac{6.0 - x}{x} = 2.0$ $x = 2.0\text{ cm}$ The location is 2.0 cm to the right of the smaller charge.
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23	C	From $t = 0$ to $t = 2.0$ min, the current in the cell = $\frac{6.0}{4.0 + (\frac{1}{3} + \frac{1}{6})^{-1} + 1.0} = \frac{6}{7} \text{ A}$ From $t = 2.0$ min to $t = 4.5$ min, the current in the cell = $\frac{6.0}{4.0 + 6.0 + 1.0} = \frac{6}{11} \text{ A}$ Using $Q = It$, total charge = $(\frac{6}{7})(2.0 \times 60) + (\frac{6}{11})(2.5 \times 60) = 184.7 \text{ C} = 180 \text{ C (to 2 s.f.)}$
24	C	The 3.0Ω and 4.0Ω resistors are in series. So effective resistance is 6.0Ω .